



INNOVATION. AUTOMATION. ANALYTICS

Exploratory Data Analysis of Crime Data (2024)

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INTRODUCTION

- The City of Los Angeles reports a large number of crimes each year. However, without proper analysis, it is difficult to identify **crime-prone areas and patterns**.
- This project focuses on analyzing **crime data from 2024** to determine **areas with higher crime intensity**, which can support decisions related to **police deployment and CCTV camera placement**.



WHY THIS PROJECT?

- This project was selected because it combines **data analysis with real-world impact**. Instead of working on a generic dataset, crime data allows the analysis to feel more realistic and purposeful.
- Understanding crime patterns can help authorities and citizens become more aware of safety issues and support better decision-making.



DATASET INFORMATION

ATTRIBUTE	DESCRIPTION
Dataset Name	Crime Data from 2020 to Present
Source	Data.gov (City of Los Angeles – LAPD)
Dataset Type	Structured (CSV format)
Time Period Considered	January 2024 – December 2024
Shape of Dataset	(134104, 29)
Purpose	To analyze crime patterns based on location, time, and crime type.

UNDERSTANDING THE DATASET

Who Collected the Data

- **Agency:** Los Angeles Police Department (LAPD)
- **LAPD** is the primary law-enforcement agency for the City of Los Angeles.
- Every reported or discovered crime is recorded in LAPD's Records Management System (**RMS**).
- This dataset is a public, anonymized extract of those records.

Study Area

- **City:** Los Angeles, California, USA
- **Latitude Range:** 33.7 – 34.5
- **Longitude Range:** -118.7 – -117.9



COLUMN DESCRIPTION

Identification

DR_NO: Unique identifier for each crime record.

Date & Time Information

Date Rptd: Date when the crime was reported to LAPD.

DATE OCC: Actual or estimated date when the crime occurred.

TIME OCC: Time of crime occurrence in 24-hour format.

Location Information

AREA: Numeric code for LAPD policing area.

AREA NAME: Name of the LAPD division (e.g., Central, Hollywood).

Rpt Dist No: Smaller reporting district used for patrol planning.

LOCATION: Block-level address of the crime (privacy protected).

Cross Street: Nearest intersecting street.

LAT / LON: Geographic coordinates of the crime location.

Crime Information

Part 1-2: Crime severity classification based on FBI UCR standards.

Crm Cd: Numeric crime classification code.

Crm Cd Desc: Text description of the crime type.

Crm Cd 1–4: Multiple crime codes if more than one crime occurred in a single incident.

Mocodes: Modus operandi codes describing suspect actions or methods.

Victim Information

Vict Age: Age of the victim.

Vict Sex: Gender of the victim (M, F, X).

Vict Descent: Coded value representing victim's race/ethnicity.

Premise & Weapon Information

Premis Cd: Numeric code describing crime location type.

Premis Desc: Text description of the premise (e.g., street, residence).

Weapon Used Cd: Code indicating weapon used, if any.

Weapon Desc: Description of the weapon used.

Case Status

Status: Short code indicating investigation status.

Status Desc: Detailed description of case status (e.g., under investigation).

TARGET VARIABLE

Part 1-2

Reason

Represents **crime severity**

Enables comparison between **serious and less severe crimes**

Useful for understanding public safety risk patterns

Part 1: Serious crimes (e.g., homicide, robbery, burglary)

Part 2: Less severe crimes (e.g., vandalism, simple assault)

INITIAL OBSERVATIONS

The dataset includes both categorical and numerical variables, with some missing values in victim and weapon-related fields. Crimes are unevenly distributed across areas and vary by time and severity, making the data suitable for exploratory analysis.

DATA CLEANING AND PREPARATION

Missing values handling

COLUMN	ACTION TAKEN
victim_age	Filled with median (in one dataset copy); in another copy labeled as Unknown to analyze bias.
victim_gender	Filled with mode (one copy); labeled as Unknown in second copy to visualize bias.
victim_ethnicity	Filled with mode (one copy); labeled as Unknown in second copy to visualize bias.
crime_method_code	Filled with mode (one copy); labeled as Unknown in second copy to visualize bias.
location_description	Filled with mode / unknown labeled
Weapon Used Cd	Dropped (>50% missing)
Crm Cd 2-4	Dropped (>50% missing)
Cross Street	Dropped (>50% missing)

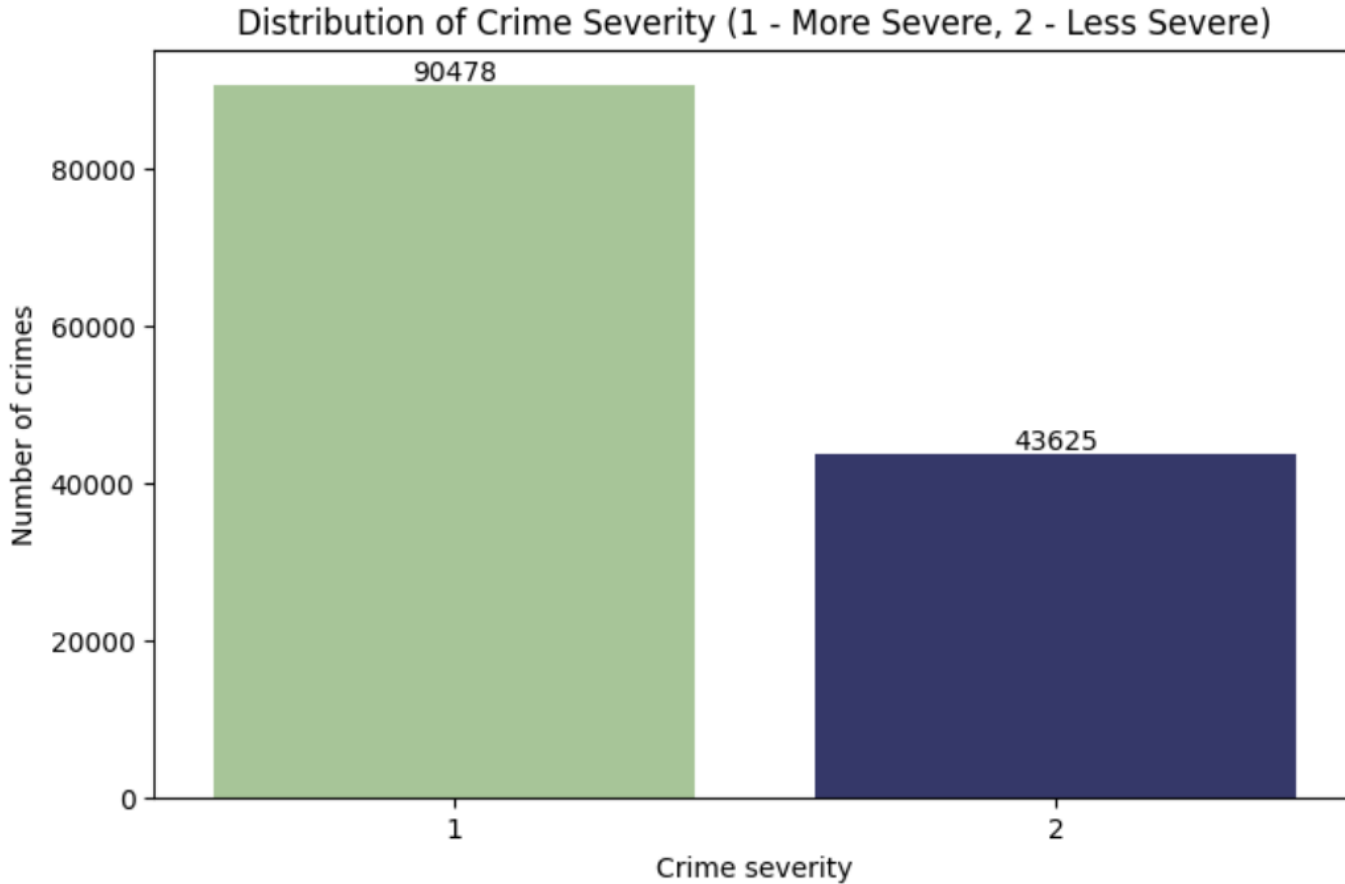
Duplicate rows

Duplicate Records: None found in the dataset → no rows removed

Outlier Treatment

Column	Outlier Count	Outlier Percentage	Treatment / Notes
victim_age	538	0.40%	Not treated as outliers; ages ≤ 100 are biologically plausible
latitude	4670	3.48%	1 row outside LA coordinates dropped; no matching info available to impute, All other coordinates were within valid LA ranges.
longitude	0	0.00%	—
reporting_delay_days	21,580	16.09%	Extreme right skew; values retained as plausible delays
area_code	0	0.00%	—
crime_district_code	0	0.00%	—
crime_severity	0	0.00%	—
crime_code	0	0.00%	—
Crm Cd 1	0	0.00%	—

CLASS IMBALANCE



Total records: **134,104**

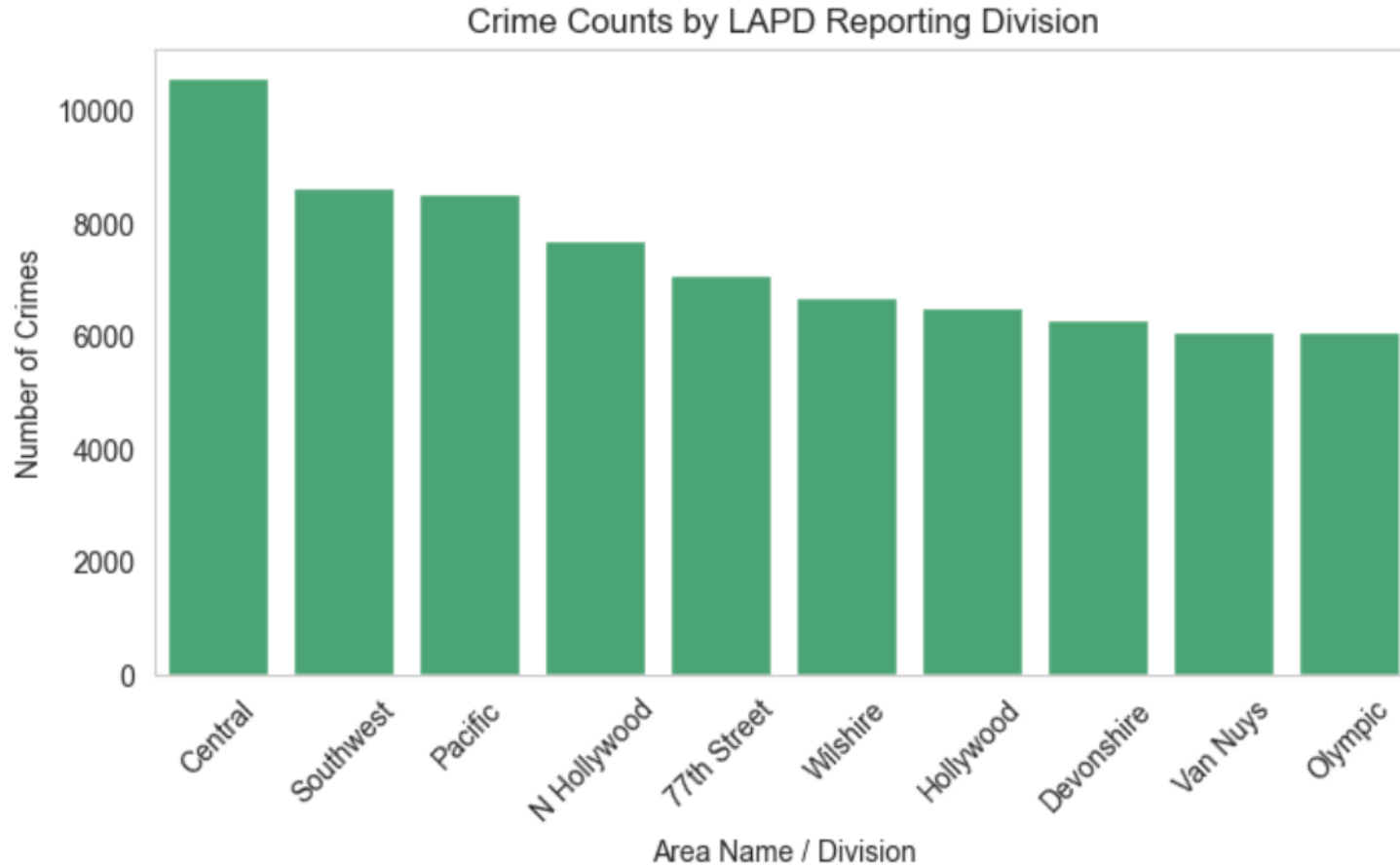
Crime severity distribution:

Part 1 (More Severe): 90,478 (~**67%**)

Part 2 (Less Severe): 43,625 (~**33%**)

The dataset shows a **significant imbalance** toward more severe crimes.

EDA – UNIVARIATE ANALYSIS

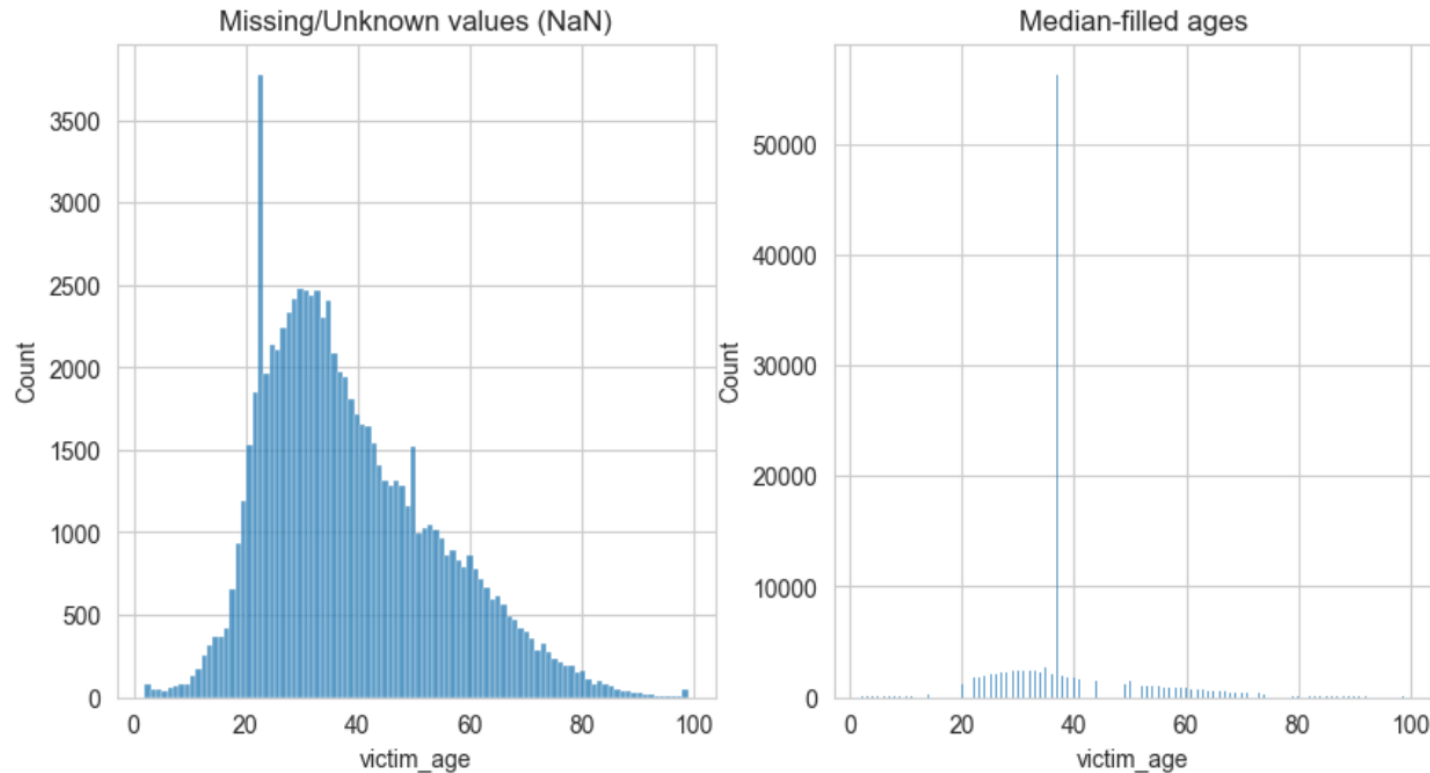


Crime is unevenly spread across Los Angeles.

The top 10 divisions account for the majority of reported incidents, with Central having the highest and Foothill the lowest.

Targeted efforts in these hotspots can improve safety.

Victim Age Distribution – Before and After Imputation



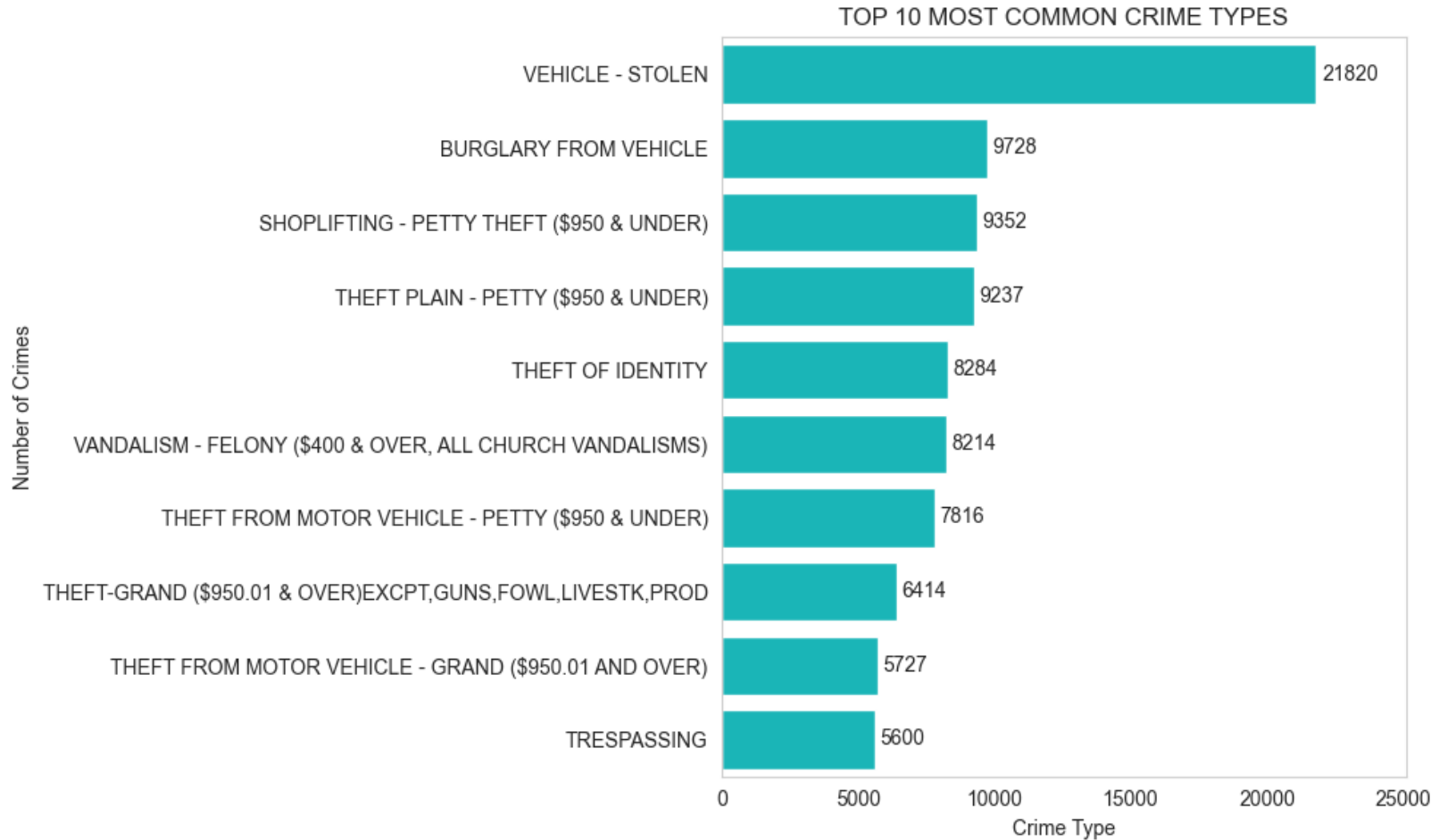
Before Imputation (NaN values): Shows original distribution with zeros/unknown ages

After Imputation (Median): Missing/invalid ages filled with median → preserves central tendency

Observation: Median imputation introduces a spike at the median value, slightly shifting the distribution

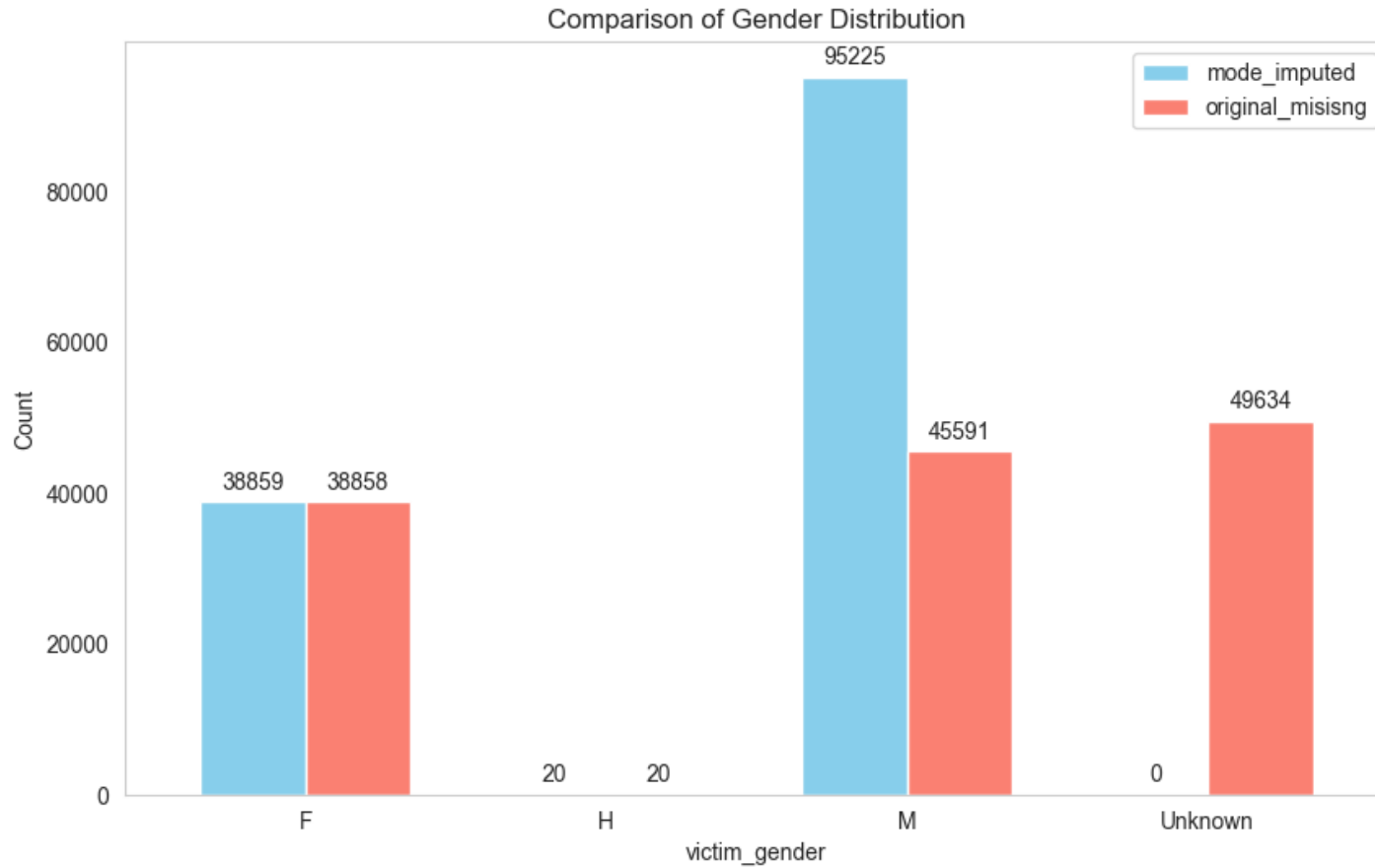
Justification: Age has no correlation with other variables, so median is simple and robust

Highest most common crime types in 2024



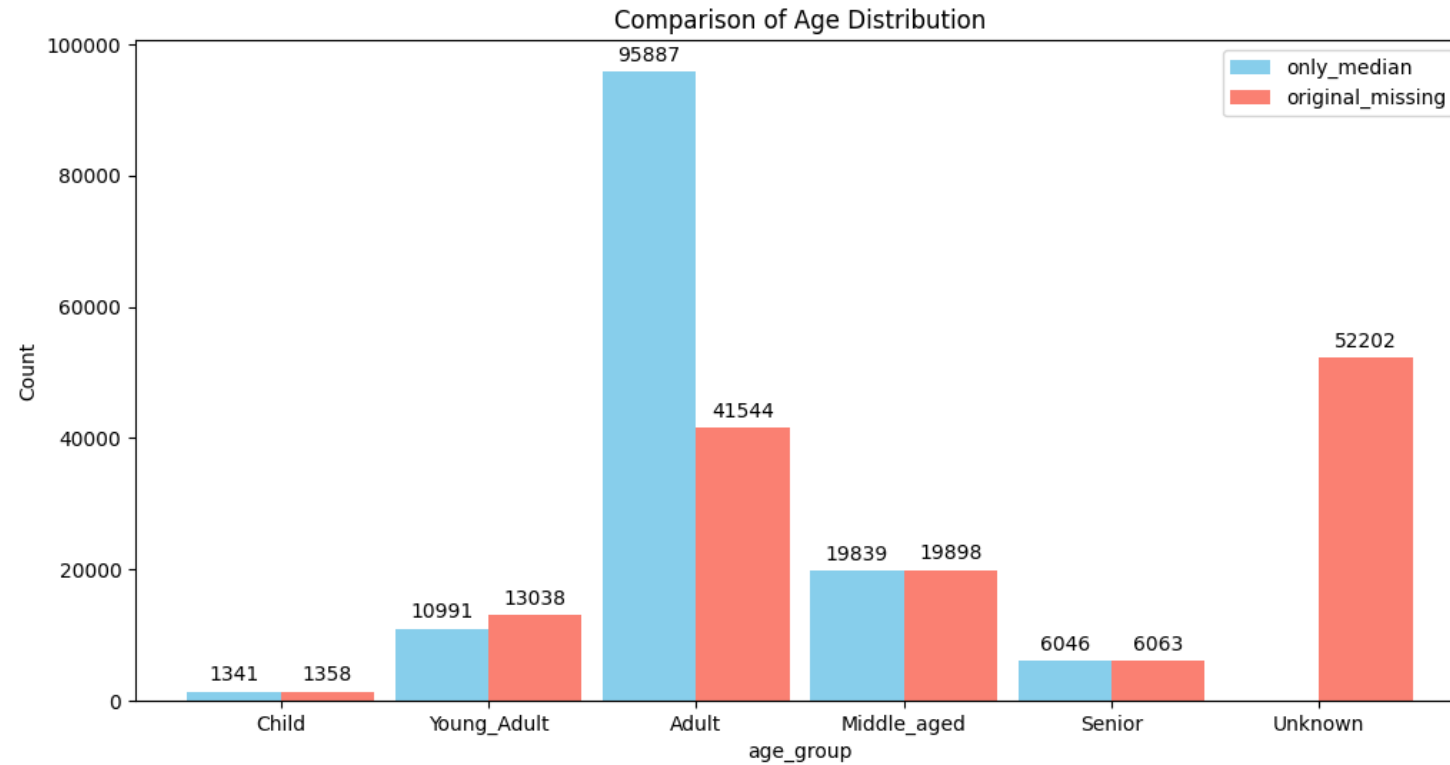
Vehicle-related crimes dominate the top 10, with 'Vehicle – Stolen' being the most common, followed by burglary from vehicles, shoplifting, and other thefts. Property crimes like vandalism and trespassing also appear, highlighting key areas for preventive efforts.

Impact of Mode Imputation on Gender



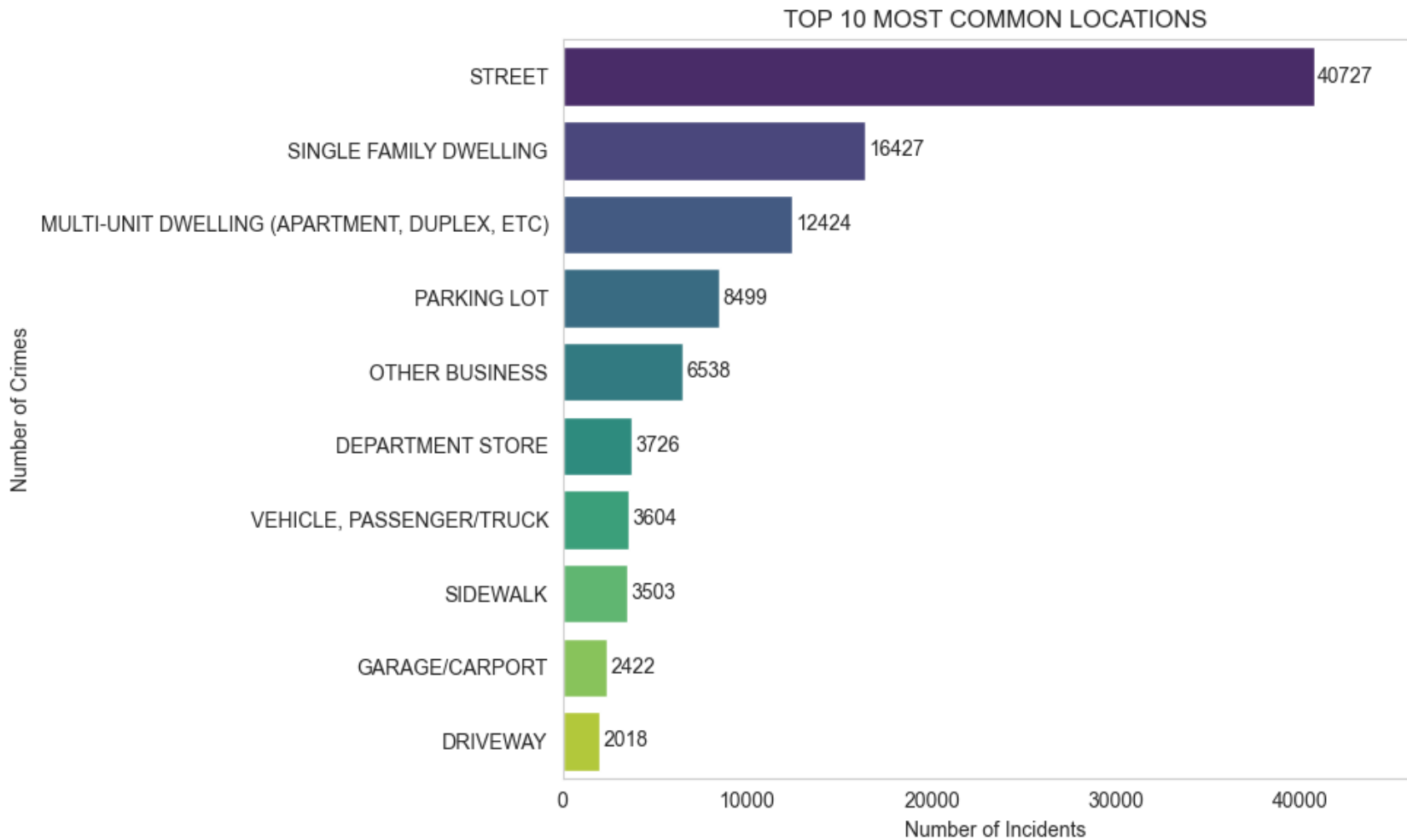
Mode imputation preserves dataset size but skews gender distribution by over-representing males and removing 'Unknown' values, potentially biasing downstream analyses

Effect of Age Imputation



Replacing missing ages with the median inflates the Adult category, distorting the true age distribution

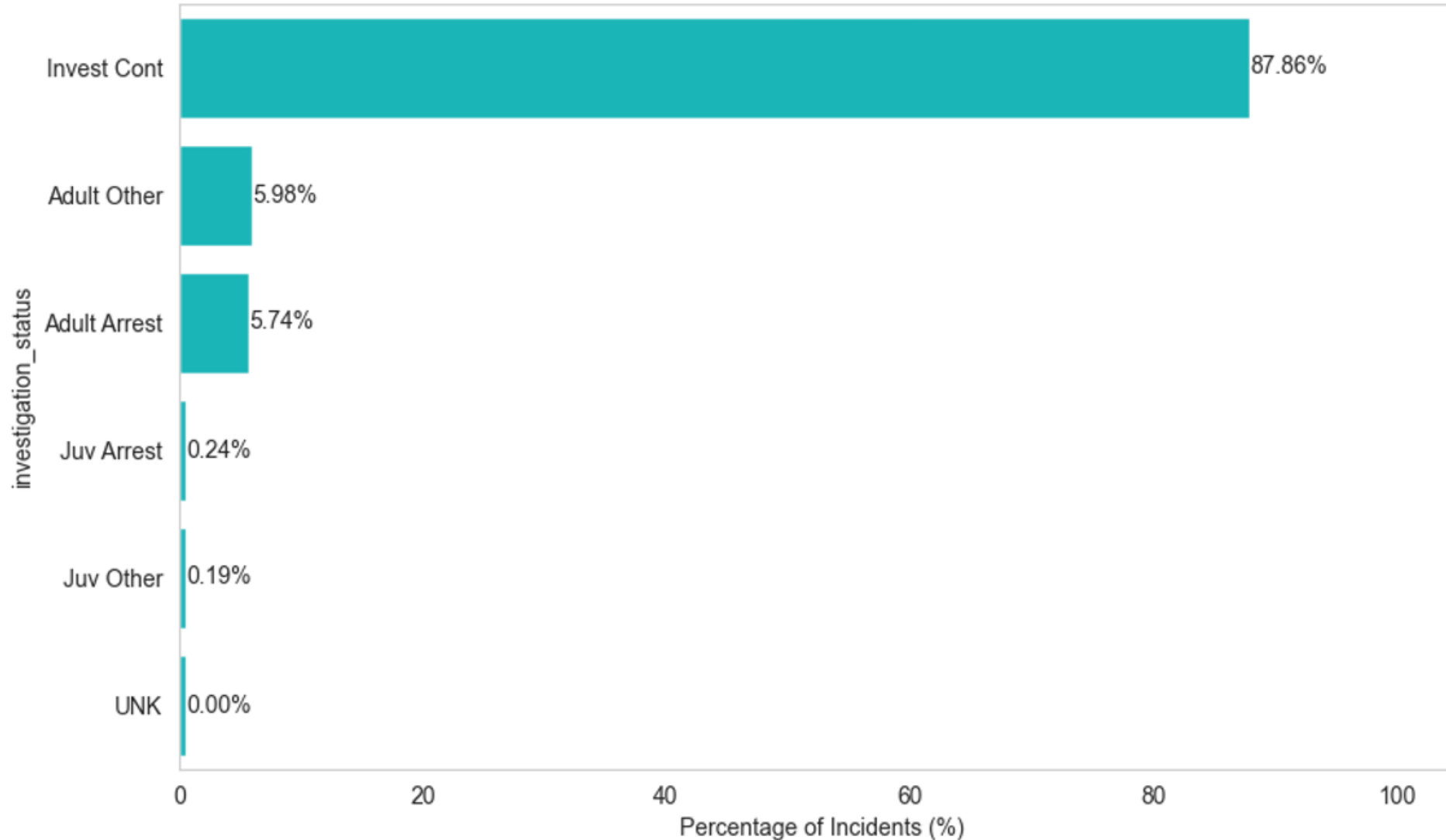
Top Crime Locations



The top 10 locations show most crimes occur on streets and in homes, indicating these areas need the highest focus for safety and patrol

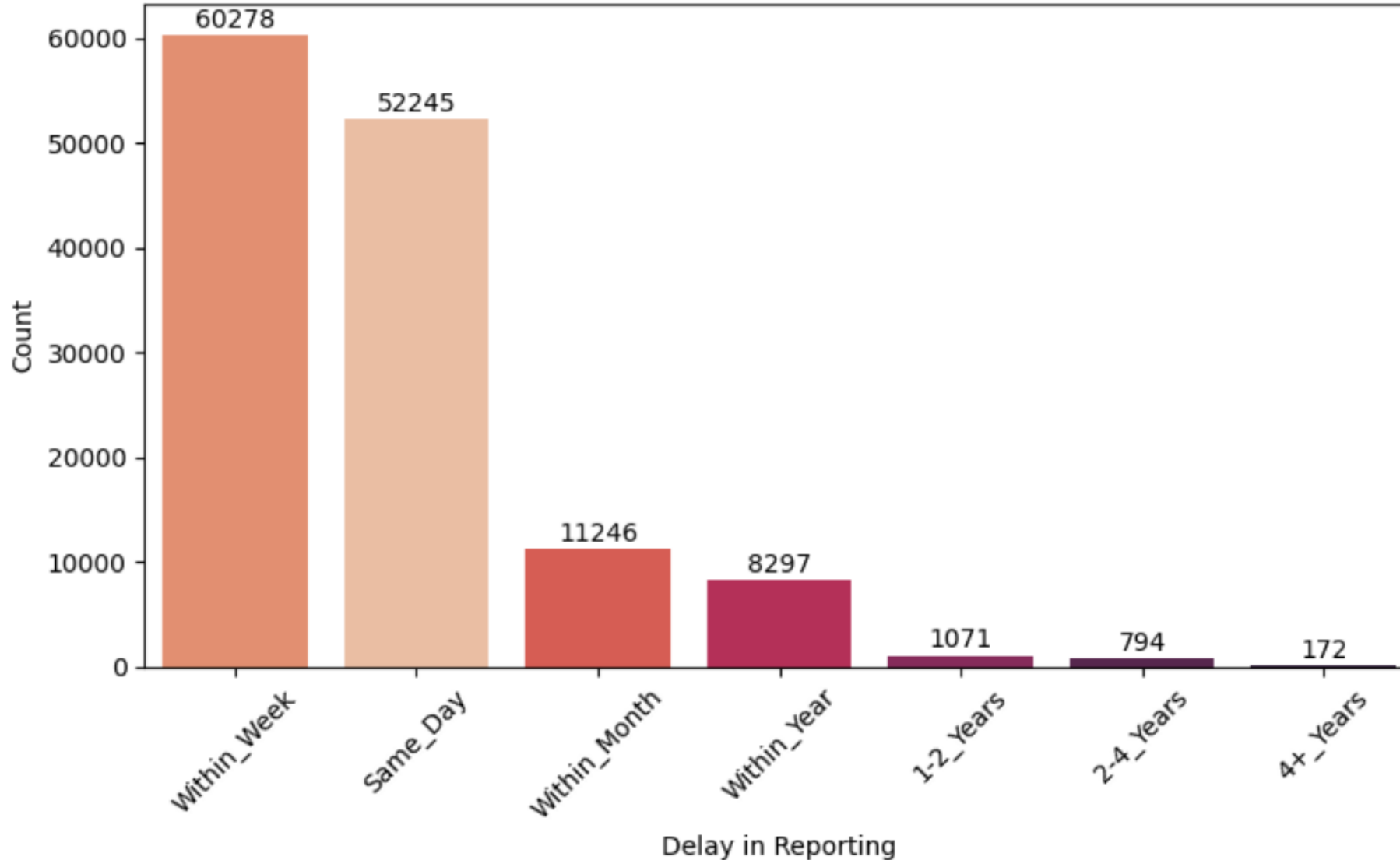
Case Outcomes

Investigation Status Distribution (Percentage)



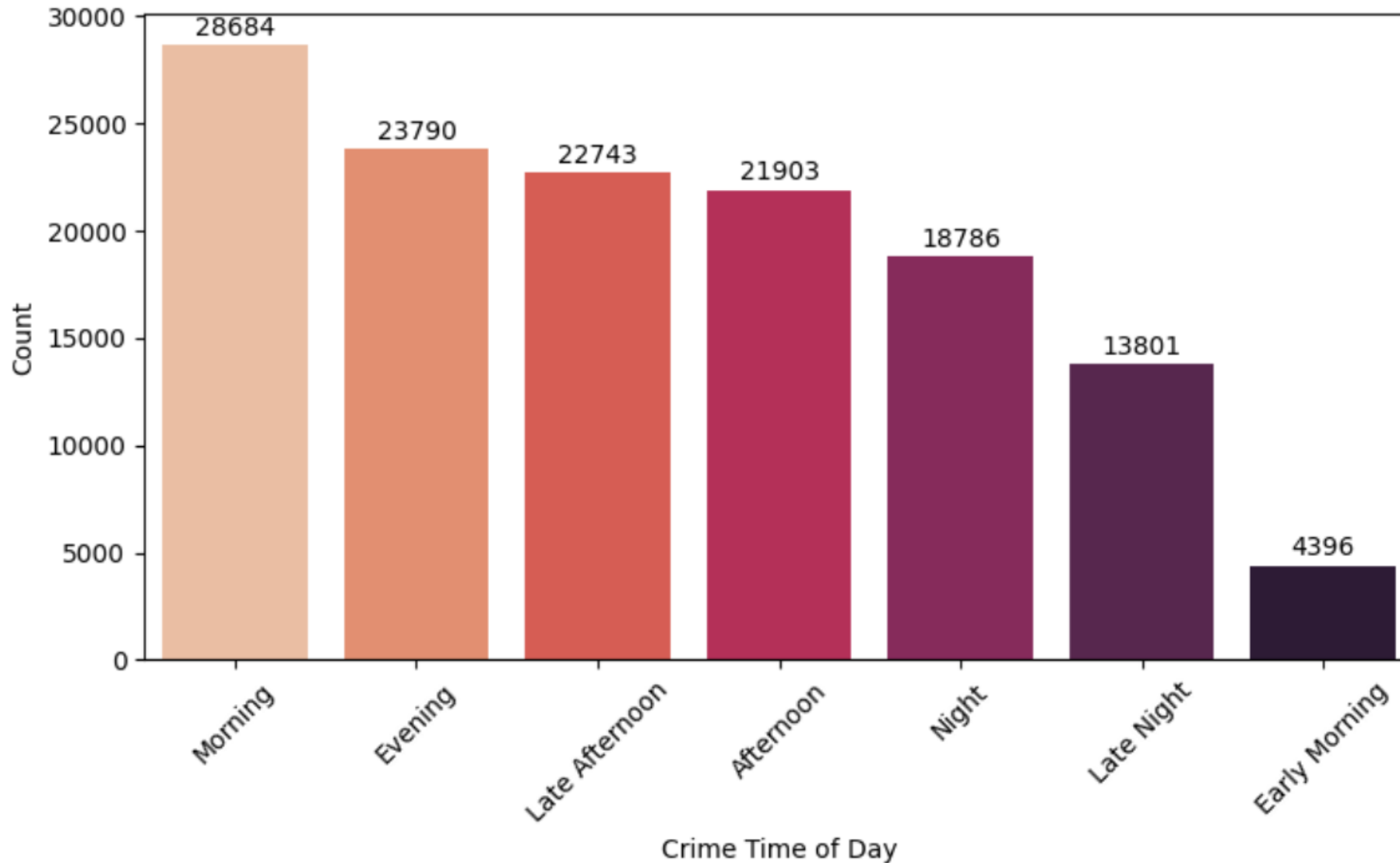
Over 90% of cases are still under investigation, while arrests make up less than 1% of incidents.

Crime Reporting Timeliness



Most crimes are reported within a day or week; long delays are rare and usually linked to fraud or identity theft.

Crime Timing



Crimes peak during daylight and early evening when activity is high, and drop significantly late at night and early morning.

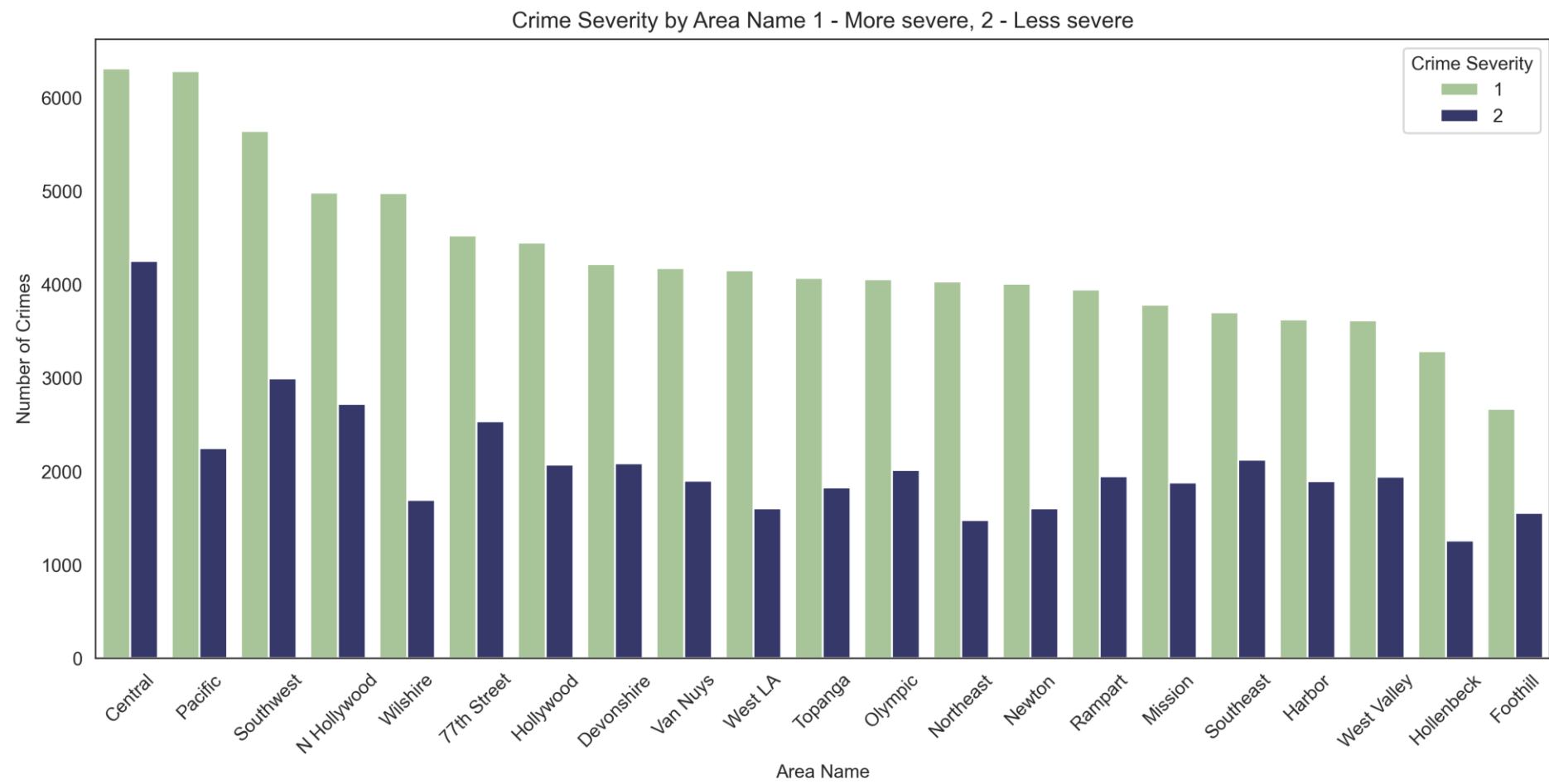
Most Common Crime Descriptions



Vehicle theft, burglary, and shoplifting are the most frequently reported crimes.

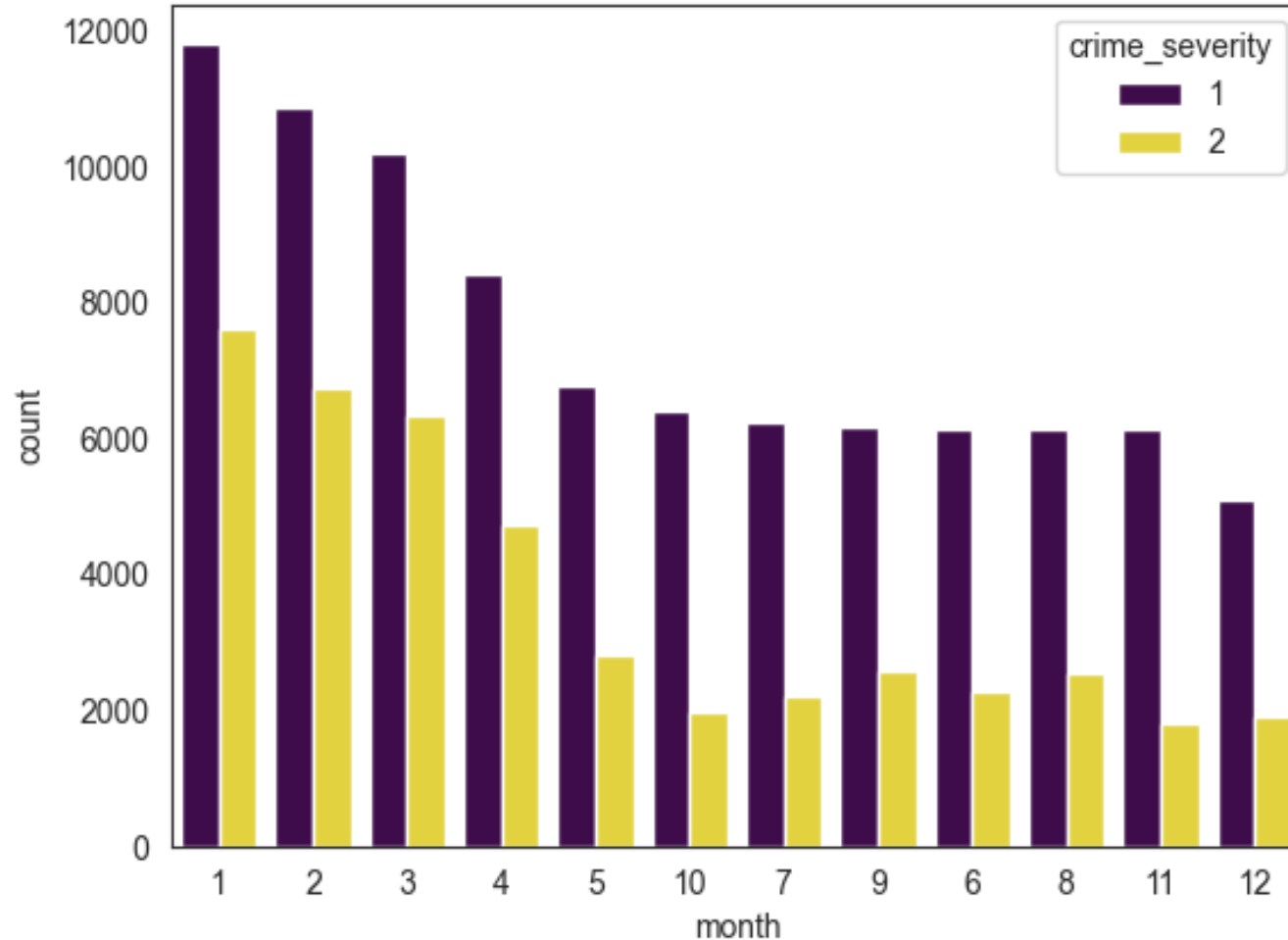
BIVARIATE ANALYSIS

Distribution of Crime Severity Across Reporting Areas (Reported in 2024)



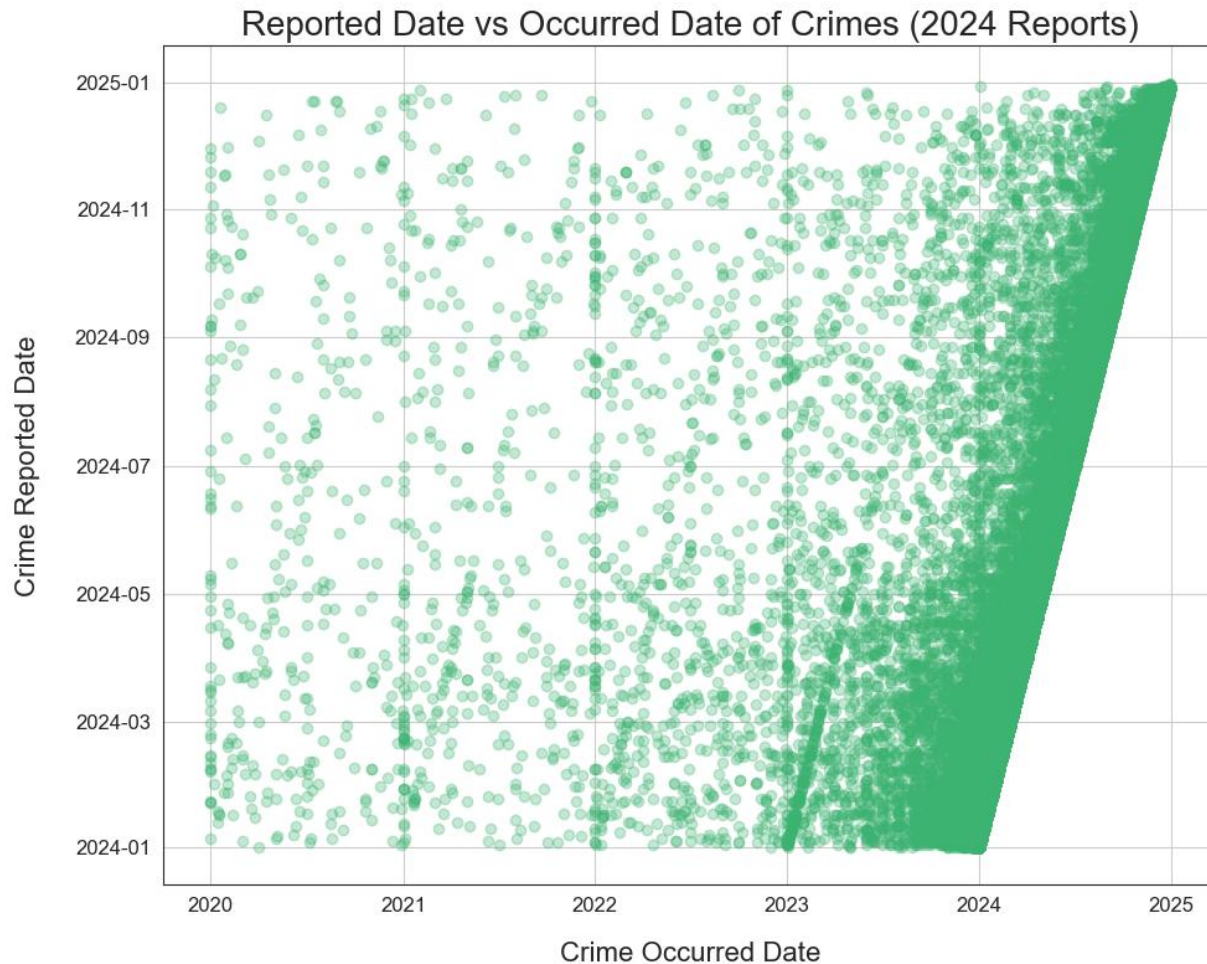
Central, Pacific, and Southwest consistently lead in Severity-1 incidents, highlighting the need for greater allocation of specialized response and investigative resources in these areas.

Crime Occurrences by Month and Severity (Occurrences up to 2024, Reported in 2024)



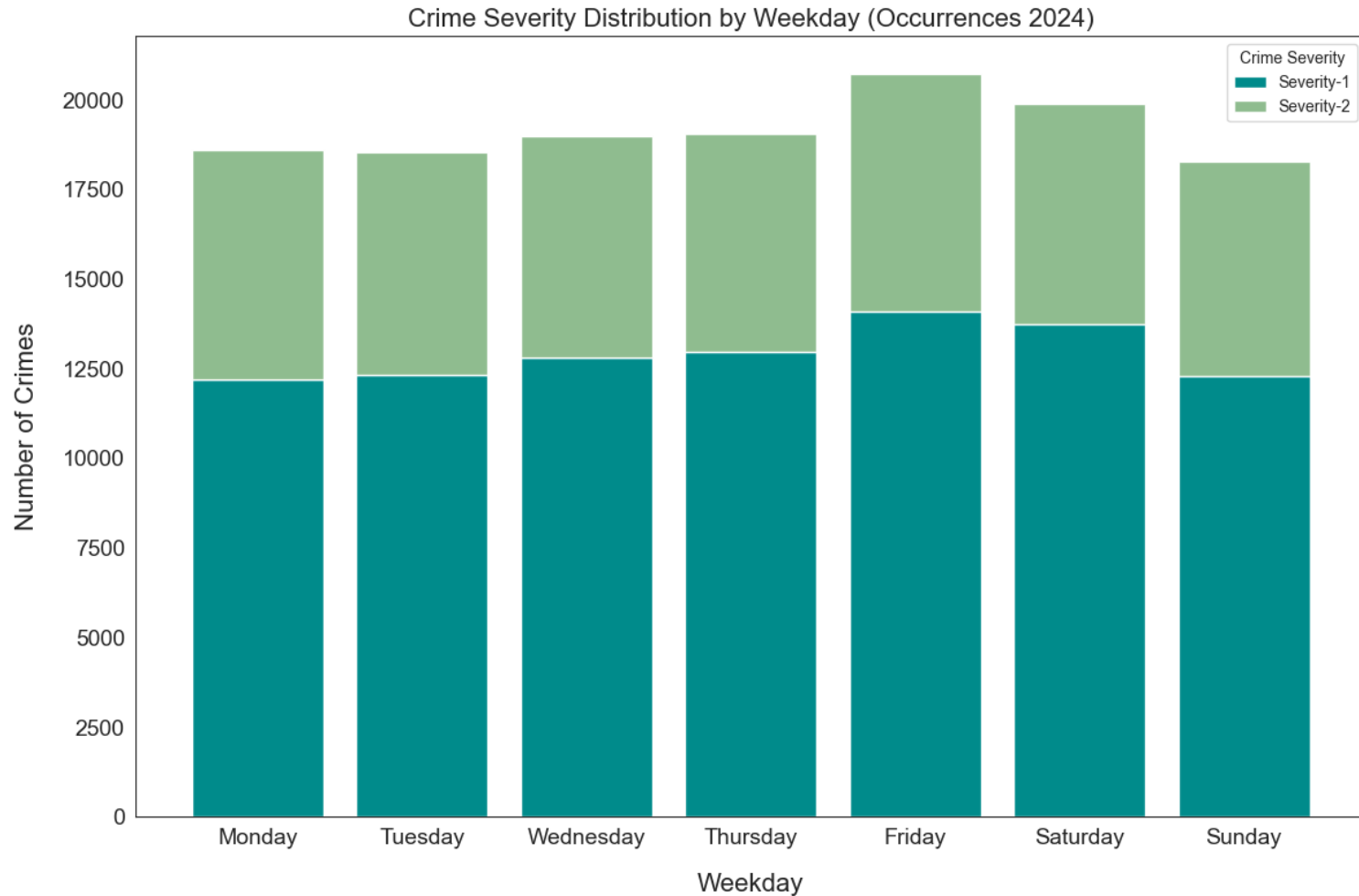
Most Severity-1 crimes occurred in **January and February**, even though they may have happened anytime between 2020–2024, and were reported in 2024. Severity-2 crimes are more evenly distributed across months. This highlights both seasonal patterns in crime occurrence and the reporting lag.

Relationship Between Occurrence and Reporting Dates of Crimes Reported in 2024”



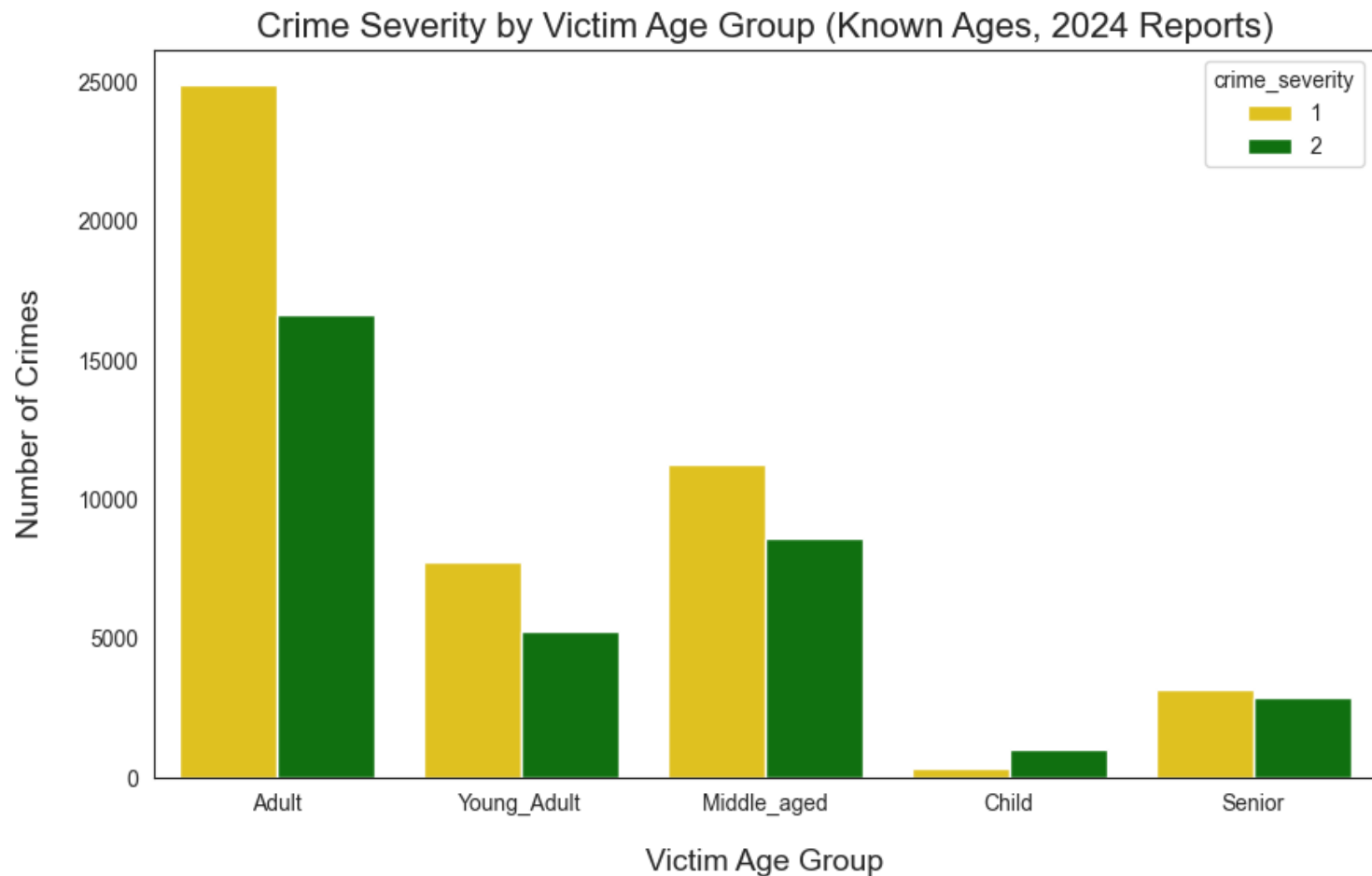
Crimes are generally reported close to their occurrence date, with some delayed reports.

Weekly Pattern of Crime Occurrence



The most severe crimes (Severity-1) peak on **Fridays and Saturdays**, indicating that the end of the week experiences the highest serious incident load. Severity-2 crimes are more evenly distributed across the week but still show a slight increase on Fridays. This trend can inform weekend-focused preventive measures and resource planning.

Crime Severity Distribution Across Victim Age Groups

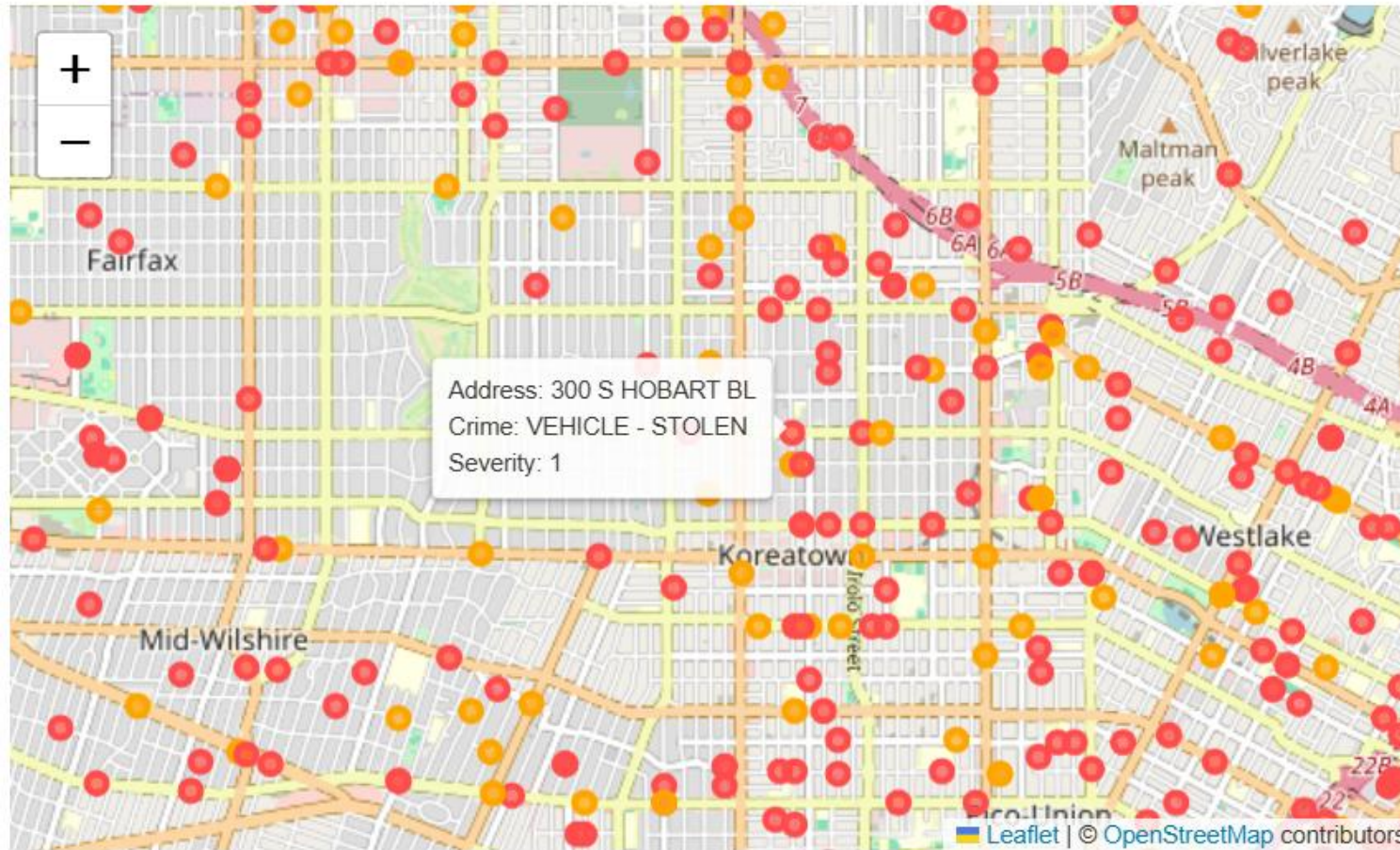


Both severity 1 and severity 2 crimes peak in the Adult age group, with severity 1 crimes being notably higher

Seniors experience relatively few crimes, with severity 1 and 2 crimes nearly balanced.

MULTIVARIATE ANALYSIS

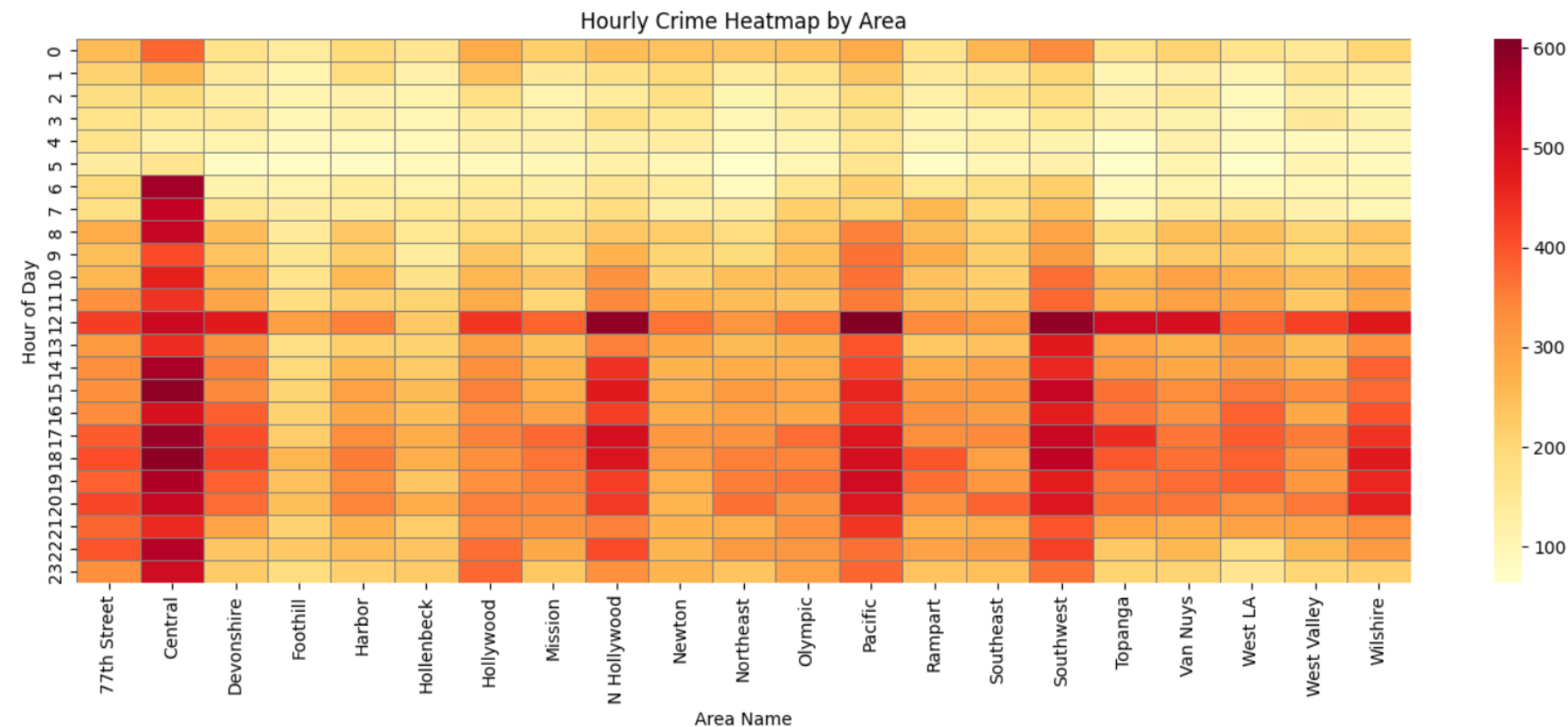
Crime Hotspots by Severity (Occurrences 2020–2024, Reported in 2024)



The map highlights spatial patterns of criminal activity. **Severity-1 crimes are concentrated in the Central, Pacific, and Southwest areas, forming clear hotspots**, while Severity-2 and Severity-3 crimes are more dispersed across the city. These hotspots indicate areas that may require increased policing and targeted preventive measures.

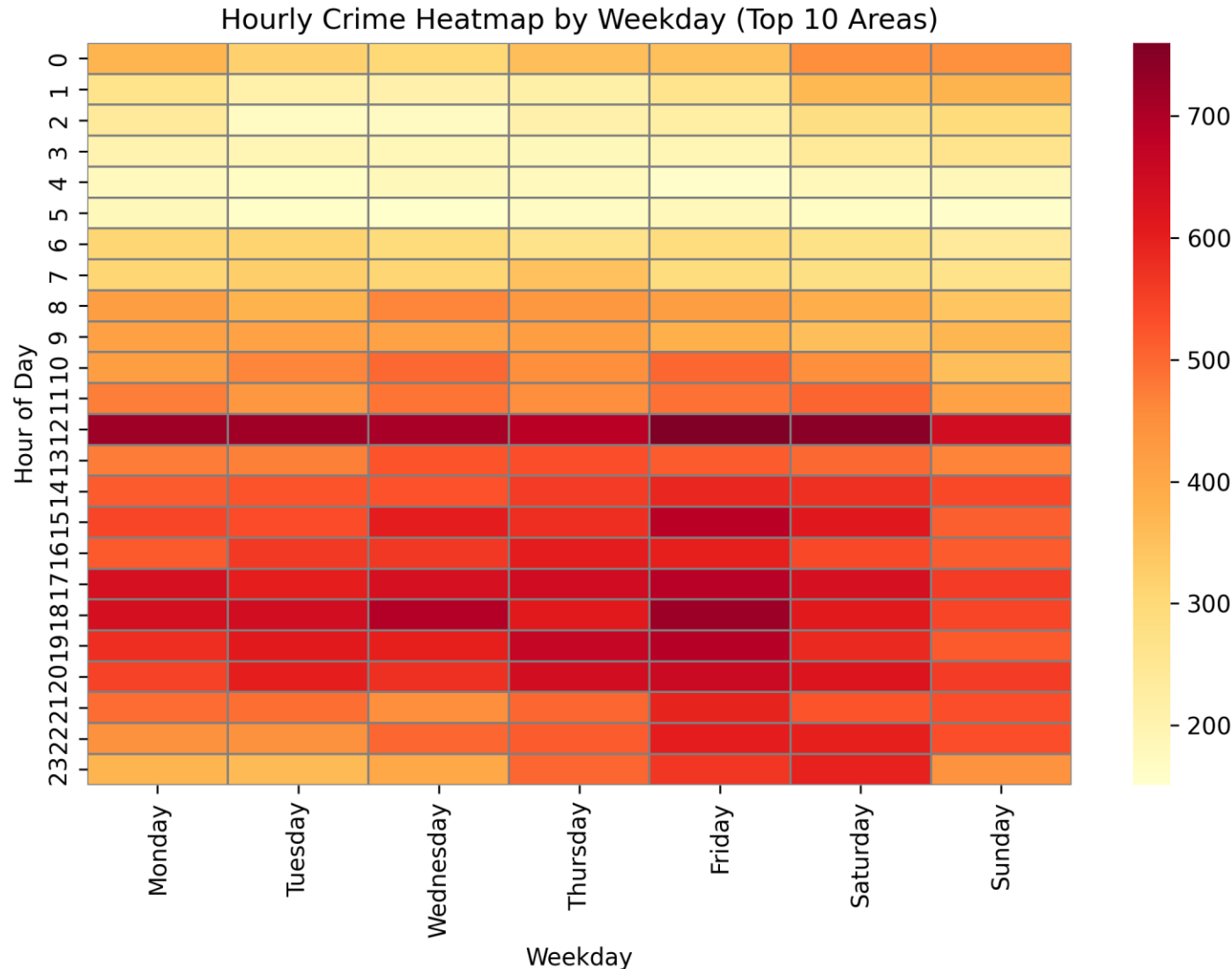
Map shows a sample of 2,000 crimes for clarity.

Hourly Crime Distribution Across Areas (Occurrences 2020–2024, Reported in 2024)



Crime activity varies by area and time of day. Most areas show **higher crime counts during late afternoon and evening hours**, while some areas like Central and Pacific exhibit consistently high activity throughout the day. This highlights both **temporal and spatial hotspots** for targeted policing.

Hourly Crime Heatmap by Weekday (Top 10 High-Crime Areas)



Crime activity varies by hour and weekday in the city's top 10 high-crime areas. **Evenings, especially on Fridays and Saturdays, experience the highest number of incidents**, while early mornings and weekdays like Monday–Wednesday see lower activity. This pattern helps identify **temporal hotspots** for strategic resource deployment.

Weekly Distribution of Crime Severity (Occurrences 2020–2024, Reported in 2024)

crime_severity	1	2
weekday		
Monday	12196	6393
Tuesday	12335	6190
Wednesday	12805	6197
Thursday	12981	6087
Friday	14107	6628
Saturday	13755	6134
Sunday	12299	5996

Severity-1 crimes peak on Fridays and Saturdays, while weekdays like Monday–Wednesday see lower counts, highlighting end-of-week hotspots for high-severity incidents.

Crime Severity by Victim Age Group (Known Ages, 2024 Reports)

crime_severity	1	2
age_group		
Adult	24919	16625
Child	340	1018
Middle_aged	11289	8609
Senior	3177	2886
Young_Adult	7774	5264

Adults are the most affected group for both Severity-1 and Severity-2 crimes, followed by Middle-aged and Young Adults, while Children and Seniors experience fewer incidents

CONCLUSION

This analysis provides a clear picture of where, when, and who is most affected by crime in the city. High-severity crimes are concentrated in specific areas and peak during evenings and weekends, while adults are disproportionately affected. Understanding these patterns enables targeted interventions, better resource allocation, and informed policy decisions to enhance public safety

FINAL INSIGHTS

- Spatial patterns:** Central, Pacific, and Southwest areas are hotspots for Severity-1 crimes.
- Temporal patterns:** Crime peaks in the **morning** with Fridays and Saturdays showing the highest activity.
- Demographics:** Adults are the most affected age group; children and seniors are less impacted.
- Reporting patterns:** Most crimes are reported shortly after occurrence, but some delays exist, highlighting opportunities for improved reporting mechanisms.
- Severity distribution:** Severity-1 crimes dominate in hotspots, while Severity-2 crimes are more evenly spread across areas.

LEARNINGS FROM THE PROJECT

- **Data handling:** Learned to manage missing values, unknown victim ages, and reporting delays effectively.
- **Visualization skills:** Applied heatmaps, scatter plots, stacked barplots, and interactive maps to uncover patterns.
- **Analytical thinking:** Connected temporal, spatial, and demographic data to generate actionable insights.
- **Presentation skills:** Learned to translate complex data into **clear visuals and concise insights** for stakeholders.
- **Client-ready insights:** Developed the ability to highlight **key trends and takeaways** without overwhelming the audience with raw data.

THANK
YOU

